

pFedDKS: Detached Knowledge Sharing for Personalized Federated Learning

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Abstract

By allowing each client to refer to the knowledge from other clients while retaining their specific characteristics, partial knowledge sharing has become one of the main approaches to realizing personalized federated learning (pFL). Representative techniques of partial knowledge sharing propose sharing the feature extractor while customizing the classifier head of the neural network. Although such methods achieve great success, the underlying principle behind them remains yet to be comprehensively understood. A fundamental problem is *whether it is really appropriate to fully share the feature extractor*. Based on the theory of neural collapse, in this paper, we demonstrate *both theoretically and empirically* that the feature extractor should be partially shared rather than fully shared. More specifically, we identify a substantial inconsistency between the fused global feature representations and expected local feature representations, and thus it is necessary to preserve partially customized layers of the feature extractor for enhancing personalized representations. Based on this discovery, we further propose a novel method called pFedDKS which detaches the shared global knowledge and customized local knowledge by providing *detached feature prototypes*. Extensive experiments on various datasets and models show that pFedDKS outperforms state-of-the-arts.

CCS Concepts

• Computer systems organization → Embedded systems.

Keywords

Federated Learning, Personalization, Knowledge Sharing

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1 Introduction

By allowing multiple clients to collaboratively train a model without directly accessing their private data, Federated Learning (FL) [1–3] recently received much attention and has been widely adopted in various applications such as healthcare [4, 5], time series data processing [6], and recommendation [7–9]. However, traditional FL suffers from a critical challenge that significantly degrades the learning performance, i.e., the not independently and identically distributed (Non-IID) user data, where the trained global model often cannot be generalized well over each client [12–14].

To deal with the above issues, employing personalized models appears to be an effective solution in FL, i.e., personalized federated learning (pFL) [15–17]. Recently, there have been many works regarding pFL, e.g., regularization-based [18, 19], meta learning based [20], clustering based [21, 22], and partial knowledge sharing (PKS) based methods [23–25]. Among all these works, PKS-based pFL methods are gaining growing attention due to their promising success. Some earlier works in this category usually share some specific layers to achieve personalization [23, 26–28], e.g., only sharing batch normalization layers and keeping the other layers customized [23]. Recently, more and more works propose sharing the full feature extractor and only keeping the classifier head customized [24, 25, 29–32]. These works typically share the feature extractor by computing the average of its parameters while aligning the local features with the global feature prototypes among clients [30–32]. Despite the significant success achieved by these methods, the underlying principle remains mysterious, i.e., it is still unknown *whether sharing the full feature extractor or only part of it is theoretically better*.

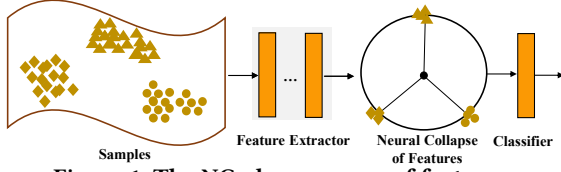


Figure 1: The NC phenomenon of features.

To solve this problem, we seek to study the underlying principles of partial knowledge sharing in pFL based upon an intriguing phenomenon that has been recently discovered in terms of the representations, termed Neural Collapse (NC) [33–35]. As shown in Figure 1, NC shows that for classification tasks, the features outputted by the extractor are expected to converge to a specific mathematical structure on the training data, where the intra-class variability of features collapses to zero for each class and the between-class feature means all collapse to the vertices of a Simplex Equiangular Tight Frame (ETF) up to scaling. Motivated by the NC phenomenon, we identify that the above question can be definitely answered by measuring whether the fused global feature representations are consistent with the expected personalized feature representations. Therefore, we leverage NC theories for analyzing feature representations in pFL and reveal an inevitable inconsistency between the fused global and the expected personalized feature representations due to the Non-IID, concluding that sharing the entire feature extractor is not optimal for personalization.

However, even though the partial extractor sharing scheme has more theoretical potential than the full scheme, recent works demonstrated that the full scheme achieves better results [32]. We identify the key reason these methods are effective lies in their use of global averaging to obtain global feature prototypes with strong discriminative ability, and these prototypes guide the training of local models, thereby fully unleashing global knowledge. However, it is still unclear how to define such a prototype scheme for the partial sharing scheme due to the inconsistency between the objectives of the shared and personalized parts, i.e., sharing knowledge and customizing knowledge. To tackle this challenge, we propose pFedDKS, a novel prototype scheme specifically tailored for the partial extractor sharing scheme. To be more specific, based on a basic partial sharing scheme that employs the shallow layers as the shared part and the remained layers as the personalized part, we utilize the NC theories to predeterminedly generate detached global and local ETF prototypes for guiding the training of the shared part and personalized part respectively. Besides, to ease the alignment between the real features and the ETF prototypes, we propose generating these prototypes in a *matching* way where the ETF vectors are computed based on the real features. Since the NC-based prototypes conform to the ETF structure, the separability of both global and local features is maximized. Experiments conducted across multiple datasets and experimental setups demonstrate that our proposed method achieves significant improvement over baselines, especially when the tasks are challenging. In particular, the upper bound improvement of our method in TinyImageNet (5.1%) is much higher than that in Fashion-MNIST (0.3%). Our contributions are:

- To the best of our knowledge, this paper is the first to dive into the mystery of partial knowledge sharing in pFL. Based on NC, we show that the feature extractor cannot be fully shared due

to the inconsistency between the learned global representations and expected personalization representations.

- We propose a novel training method, namely, pFedDKS, that leverages NC theories to provide detached feature prototypes tailored for the partial extractor sharing scheme. In this way, the global features are aligned for ease of sharing global knowledge while maximizing the separability of personalized features.
- We propose generating the global and local ETF prototypes in a *matching* way by calculating the ETF vectors based on the real features obtained in a warmup manner.
- We compare pFedDKS with many advanced partial sharing based methods by conducting extensive experiments over various models and datasets, where the results show that our method can improve the accuracy by up to 5.1%.

2 Related Work

Personalized Federated Learning. Partial sharing has become a key approach for personalized federated learning [27, 28, 36, 37]. Earlier studies focused on sharing specific layers, such as batch normalization [23]. Recent works propose sharing the full feature extractor and customizing only the classifier head [24, 25, 31, 38, 39]. Examples include Fed-RoD [24], which shares the full feature extractor but uses different softmax functions for global and local learning. FedBABU [25] maintains a global classifier while fine-tuning locally. kNN-Per [29] combines a global model with local kNN classifiers. LG-FedAvg [40] learns the entire network locally but aggregates only top layers. FedRep [31] averages feature extractor parameters, while FedProto [30] aligns feature representations among clients. FedPAC [32] combines both averaging and alignment. Unlike these, we focus on demystifying partial feature extractor sharing and propose a novel harmonized prototype scheme. Some studies apply NC theories to FL [41, 42], but they mainly aim to train global models without considering prototypes, which is our main contribution.

Neural Collapse. Papayan et al. [33] first observed the neural collapse phenomenon in balanced dataset training. Though intuitive, its cause is not fully understood. Research shows this can occur under any loss function when treating last layer features/classifiers as independent variables [34, 43, 44]. Other studies explore its link to deep network generalization [45], and feature variability collapse from shallow to deep layers improves test performance [46]. Imbalanced data research indicates specific loss functions can induce neural collapse in such settings [47, 48]. This paper proposes using these theories to dive deep into the pFL.

3 Preliminaries

3.1 Problem Formulation

We aim to collaboratively train personalized models for a set of clients in FL. The goal is to learn a function that maps every input data to the correct class out of C possible options. We assume that there are N clients, and each client n can only access its private dataset $\mathbb{D}_n := \{x_{n,i}, y_{n,i}\}$, where x_i is the i -th input data sample, $y_{n,i}$ is the corresponding label, and there are total C classes. The number of data samples in dataset $\mathbb{D}_n$ is denoted by D_n . $\mathbb{D} = \{\mathbb{D}_1, \mathbb{D}_2, \dots, \mathbb{D}_N\}$, $D = \sum_{n=1}^N D_n$. In personalized FL, the goal of the learning system is to learn N personalized models

$\mathbf{W} = \{\mathbf{w}_1, \dots, \mathbf{w}_N\}$ that minimizes the average empirical loss:

$$\min_{\mathbf{W}} \mathcal{L}(\mathbf{W}) := \sum_{n=1}^N \frac{D_n}{D} \mathcal{L}_n(\mathbf{w}_n), \quad (1)$$

$$\text{where } \mathcal{L}_n(\mathbf{w}_n) = \frac{1}{D_n} \sum_{(x_{n,i}, y_{n,i}) \in \mathbb{D}_n} \mathcal{L}_{CE}(\mathbf{w}_n; x_{n,i}, y_{n,i}),$$

where $\mathcal{L}_n(\mathbf{w})$ is the n -th client's local loss function that measures the local empirical risk over the private dataset $\mathbb{D}_n$ and $\mathcal{L}_{CE}$ is the cross-entropy loss function that measures the difference between the predicted values and the ground truth labels.

3.2 Neural Collapse

DEFINITION 1. (Simplex Equiangular Tight Frame). Considering C vectors $\mathbf{e}_i \in \mathbb{R}^d$, $i = 1, \dots, C$ and $C - 1 \leq d$, we say that they construct a simplex equiangular tight frame if:

$$\mathbf{E} = \sqrt{\frac{C}{C-1}} \mathbf{U}(\mathbf{I}_C - \frac{1}{C} \mathbf{1}_C \mathbf{1}_C^T), \quad (2)$$

where $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_C] \in \mathbb{R}^{d \times C}$, $\mathbf{U} \in \mathbb{R}^{d \times C}$ allows a rotation, satisfying $\mathbf{U}^T \mathbf{U} = \mathbf{I}_C$, $\mathbf{I}_C$ is the identity matrix, and $\mathbf{1}_C$ is an all-ones vector.

Define the feature outputted by the l -th layer of the neural network as $\mathbf{h}_{c,i}^l$ for any sample $x_{c,i}$ with the label c . Considering there are total L layers, the feature generated from the feature extractor can be denoted as $\mathbf{h}_{c,i}^{L-1}$. Now, we first present the NC phenomenon [33, 34] on the balance dataset where the samples number of each class is the same, which reveals that the features outputted by the feature extractor converge to a specific mathematical structure:

NC1: Within-class variability collapse. For each class, their features will collapse into the means, i.e., $\Sigma_H^c \rightarrow 0$, where $\Sigma_H^c = \text{Avg}_i (\mathbf{h}_{c,i}^{L-1} - \bar{\mathbf{h}}_c^{L-1})(\mathbf{h}_{c,i}^{L-1} - \bar{\mathbf{h}}_c^{L-1})^T$, with $\bar{\mathbf{h}}_c^{L-1} = \text{Avg}_i \mathbf{h}_{c,i}^{L-1}$;

NC2: Between-class features converge to ETF. For each class c , the normalized feature means $\hat{\mathbf{h}}_c^{L-1} = \frac{\mathbf{h}_c^{L-1} - \bar{\mathbf{h}}_c^{L-1}}{\|\mathbf{h}_c^{L-1} - \bar{\mathbf{h}}_c^{L-1}\|}$, $1 \leq c \leq C$ satisfy (2), where $\bar{\mathbf{h}}_c^{L-1} = \text{Avg}_{c,i} (\mathbf{h}_{c,i}^{L-1})$ is the global mean;

NC3: Self-duality. The normalized feature means $\hat{\mathbf{h}}_c^{L-1}$ are aligned with their corresponding classifier weights, i.e., $\hat{\mathbf{h}}_c^{L-1} - \frac{\mathbf{w}_c^L}{\|\mathbf{w}_c^L\|} \rightarrow 0$, where $\mathbf{w}_c^L$ is the classifier weight of class c ;

NC4: Simplification to nearest class center. The features can be simplified to the nearest class centers, i.e., $\arg \max_c \langle \mathbf{h}^{L-1}, \mathbf{w}_c^L \rangle = \arg \min_c \|\mathbf{h}^{L-1} - \bar{\mathbf{h}}_c^{L-1}\|$.

The NC phenomenon applies to unbalanced datasets, where the sample numbers of each class are not equivalent, and exhibits similarities with balanced datasets, except for NC2. As highlighted in [49], the following holds true:

Minority collapse. Considering there are C_A majority classes, each contains D^A training samples ($D^1 = \dots = D^{C_A} = D^A$), and $C_B = C - C_A$ minority classes, each contains D^B samples ($D^{C_A+1} = \dots = D^{C_A+C_B} = D^B$). Then, by training the model using the cross-entropy loss, defining the ratio $R = D^A/D^B$, the weight of minority classifiers converge to a single vector, i.e., $\lim_{R \rightarrow \infty} \mathbf{w}_c^L - \mathbf{w}_{c'}^L = 0$, for all $C_A < c \leq c' \leq C$.

It should be noted that while the aforementioned claims have not been proven for general learning paradigms, the layer-peeled models (LPM) [43, 44, 49], used to study the properties of the remaining parts by progressively "peeling away" certain layers of the DNN,

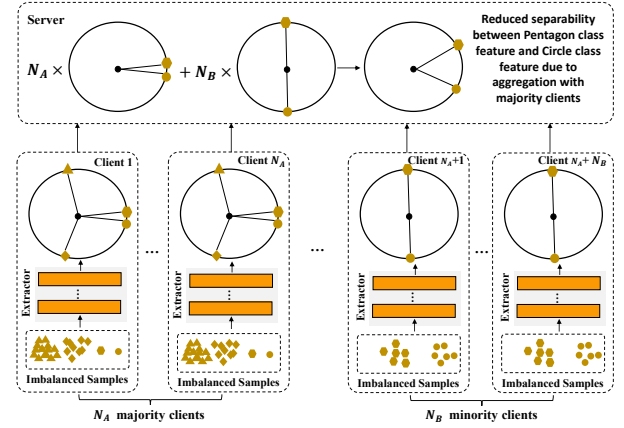


Figure 2: The interference of global knowledge.

offer theoretical guarantees. LPMs inherit the learning behaviors of extractor features and classifiers from deep neural networks [47], and can provide valuable understandings.

4 Limits of sharing full extractor

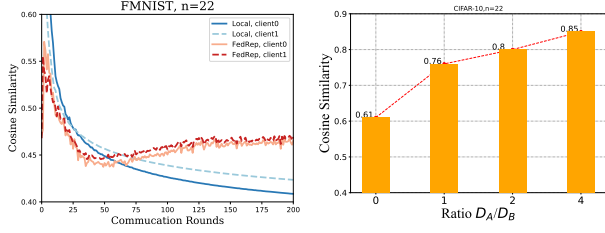
This section shows that sharing an entire extractor is suboptimal for pFL because of the inconsistency between global features and expected personalized features. We achieve this by showing that the separability of converged global features is smaller than that of optimal personalized features.

4.1 Interference across Clients

Although the knowledge from other clients can enhance the generalization capabilities of the given client, it conversely also constrains the client from reaching its optimal performance, a phenomenon particularly salient in the context of FL characterized by data heterogeneity. In the following, we show this from three aspects, i.e., case study, empirical demonstration, and theoretical analysis.

Case study. We consider the global dataset comprises only 4 classes. As illustrated in Figure 2, there are N_A clients (termed majority clients) possessing data from all 4 classes, but the quantities of pentagon and circle classes significantly trail those of the other 2 classes. Conversely, the remaining N_B clients (termed minority clients) exclusively contain data from the pentagon and circle classes. Under these circumstances, following local training, due to the limited amount of data for the pentagon and circle classes, the separability between their feature means in the models trained by the majority clients, as per [49], will be relatively low. On the other hand, minority clients' models exhibit higher separability between the feature means of the pentagon and circle classes. Should an averaging approach similar to that in [31] be employed, where more details are in Appendix A.1, the global separability between the feature means of the pentagon and circle classes would be diminished by the influence of majority clients, resulting in a value lower than the separability between these classes in minority clients. This issue becomes particularly acute when $N_A \gg N_B$, indicating that sharing a full feature extractor may not be beneficial for the personalization of clients with distinctive characteristics.

Empirical demonstration. To verify our statement, we conducted an experiment using a similar setting to the above case. The experiment is conducted in an Fashion-MNIST (simplified as FMNIST)



(a) Majority clients have interfered with the separability of minority clients' local classes. (b) Cosine similarity between the local class features within minority clients using FedAvg.

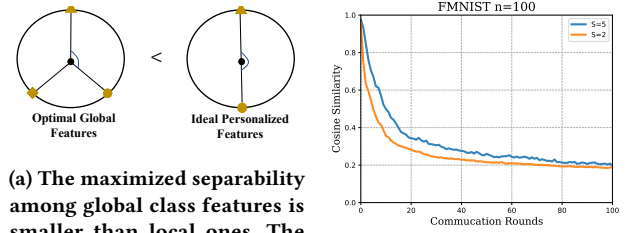
Figure 3: Separability between classes.

dataset with 22 clients. Clients 2, ..., 21 are configured as the majority clients and each of them contains 10 classes. The number of samples in class 2 : 9 is much larger than the class 0 : 1. Clients 0 and client 1 are configured as minor clients and each of them only contains two classes, i.e., 0 and 1. Based on this setup, we run two methods to compare their separability: 1) each client only trains the feature extractor using its local data; 2) all clients collaboratively share the full feature extractor and prototypes, i.e., running FedRep [31]. We measure the separability between features of different classes by computing their cosine similarity. Smaller cosine similarity means larger separability between classes. The results are shown in Figure 3a. In this figure, "Local, client 0" denotes the cosine similarity between the features of local classes 0 and 1 in client 0 under the training setting of only using the local data without sharing parameters with other clients, and "FedRep, client 0" denotes the cosine similarity between the features of local classes 0 and 1 under the training setting of using the global data, i.e., sharing parameters and prototypes of client 0 with other clients. As can be seen, the similarity between feature means of local training is smaller than that of shared training, indicating that the separability between local features of different local classes obtained by using only local data is larger than that using global data.

Theoretical analysis. Consider N_A majority clients that contain C_A majority classes, each with D^A samples such that $D^1 = \dots = D^{C_A} = D^A$, and $C_B = C - C_A$ minority classes which have D^B samples $D^{A+1} = \dots = D^{A+C_B} = D^B$. Additionally, the remaining $N_B = N - N_A$ minority clients each contain the same C_B minority classes where each has D^M samples. We have the following analysis.

PROPOSITION 1. Consider the number of major-class samples is much larger than that of minor class, i.e., the ratio $D^A/D^B \rightarrow \infty$. Suppose the total sample size of minor-class samples in majority clients is much larger than that of minority clients, i.e., $\frac{N_A D^B}{N_B D^M} \rightarrow \infty$. Then, the separability between the minor-class global feature means is approaching zero, i.e., $\frac{1}{N} \sum_{n=1}^N \bar{\mathbf{h}}_{n,c}^{L-1} - \bar{\mathbf{h}}_A^{L-1} \rightarrow 0$ for all $C_A < c \leq C$.

Proof is in Appendix A.2. Proposition 1 shows global knowledge cannot always promote the learning of personalized representations and may even hinder their improvement in an extreme case. Indeed, Figure 3b also verifies this by show that the angle between minority classes is approaching 0 (i.e., the cosine value approaching 1) when the ratio D_A/D_B becomes higher (from 0 to 4).



(a) The maximized separability among global class features is smaller than local ones. The maximized separability among 3 global classes is $\frac{2\pi}{3}$ but it is π between 2 local classes. (b) The separability between the optimal global features is smaller than local features.

Figure 4: The limitation of optimal global knowledge.

4.2 Limitation of Optimal Global Knowledge

Indeed, even if data heterogeneity exerts no direct influence on the knowledge fusion process, the optimally aggregated global knowledge still fails to reach the individual performance pinnacle anticipated for each client. We also verify this in three aspects.

Case study. We first show that the separability of converged global features is inferior to that of anticipated personalized features from an intuitive perspective. Consider a case where there are total 3 classes across all clients and 2 classes in each specific client. Assuming that the global features of the total 3 classes are ideally trained, where the separability of different class features is maximized and the between-class means of global features satisfy the ETF condition. Then, the angle between any two global feature means is equivalent to $\frac{2\pi}{3}$. However, for expected personalized features, the goal is to maximize the separability of 2 local classes, and the maximum angle is π . As illustrated in Figure 4a, the angle between optimal global feature means is $\frac{\pi}{3}$ less than local feature means.

Empirical demonstration. We then make further verification in an empirical view. We consider there are a total of 5 global classes across all clients and 2 local classes in each client. We train ResNet18 on FMNIST. To ensure the global model remains unaffected by Non-IID characteristics and achieves optimal convergence performance, we aggregate data of five classes of all clients and train the model using a centralized approach, subsequently computing the average of separability (i.e., cosine similarity) between feature means of each two-class pairs, as depicted by the $s = 5$ curve in Figure 4b. Analogously, we also collect data from the same two classes across all clients, centrally train the model, and compute the separability between class feature means, with the results represented by the $s = 2$ curve in the figure. It can be seen that the cosine value of 2-class personalized features is smaller than the 5-class global features, indicating larger separability of 2-class personalized features.

Theoretical analysis. Considering there are C_n local classes in the local data of each client n and C classes in the global data of all clients, then we have the following formal statement.

PROPOSITION 2. We consider the angle between the feature means of two classes is defined as their separability, then for local classes in a client, the separability between their optimal personalized features is greater than their optimal global features, and the difference between them is at least $\arccos(-\frac{1}{C_n-1}) - \arccos(-\frac{1}{C-1})$.

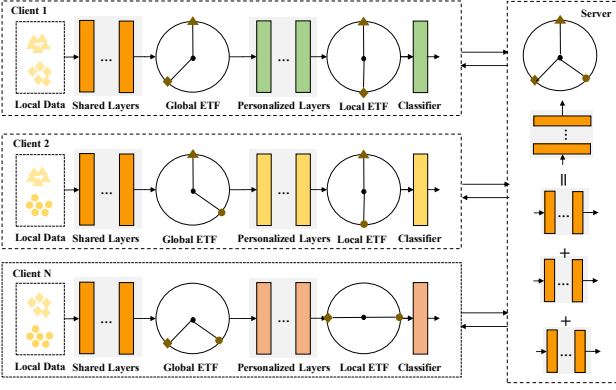


Figure 5: pFedDKS framework. Regularized by the detached global/local ETF features, each client alternatively updates the shared and the personalized parts. Server receives shared layers from all clients and computes the average.

Proof can be found in Appendix A.3. Since the number of local classes in client n is no larger than the total number of global classes, i.e., $C_n \leq C$, it always holds that $\arccos(-\frac{1}{C-1}) \leq \arccos(-\frac{1}{C_n-1})$. To this end, Proposition 2 shows that fully sharing the feature extractor cannot achieve optimal features for the local classes. It is necessary to preserve partial layers of the feature extractor to maximize the separability among local classes.

5 Methodology and Analysis

Above analysis presents the necessary of preserving personalized layers of the extractor to transform the global features into personalized ones to maximize the separability. Thus, we propose the method, pFedDKS, as shown in Figure 5. Given the variety of partial extractor sharing schemes [50], we opt for the simplest: sharing shallow layers while customizing deep layers. Our approach can integrate with other partial sharing strategies. We divide the feature extractor into a shared segment, L_s shallow layers near the input, and a customized segment, the remaining $L - L_s$ layers (including $L - 1 - L_s$ deep extractor layers and 1 classifier layer). Applying our detached prototype scheme aids in training both network segments efficiently. This scheme utilizes two types of prototypes—global ETF prototypes for aligning shared features across clients and local ETF prototypes for maximizing the separability of personalized features, detailed in the following sections.

5.1 Align Global Features with Global ETF

To maximize separability of global features, we utilize a shared partial feature extractor and align its outputs $\mathbf{h}_{n,i}^{L_s} \in \mathbb{R}^{d_s}$ with predefined global ETF prototypes. For ease of alignment, we require that the predefined ETF prototypes are close to original features, which we call *matching* ETF prototypes. To achieve this, all clients first compute the global mean of features $\bar{\mathbf{h}}^{L_s}$ and then calculate the length average of normalized features $\text{Avg}_i \|\mathbf{h}_{n,i}^{L_s} - \bar{\mathbf{h}}^{L_s}\|$, after a *warmup* phase. Next, each client locally generates vectors $\mathbf{e}_c^s \in \mathbb{R}^{d_s}$, $c = 1, \dots, C$ using the definition (2) with the same random seed. Finally, each client constructs ETF prototypes using NC2:

$$\tilde{\mathbf{h}}_c^{L_s} = \text{Avg}_{n,i} \|\mathbf{h}_{n,i}^{L_s} - \bar{\mathbf{h}}^{L_s}\| \mathbf{e}_c^s + \bar{\mathbf{h}}^{L_s} \quad (3)$$

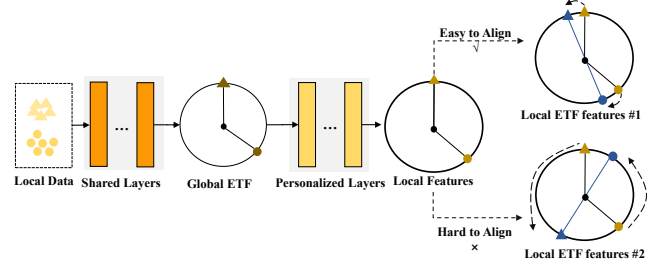


Figure 6: Illustration of aligning local features (in orange) to local ETF vectors (in blue). Top ETF features (#1) are closer to the local features than the bottom ones (#2) regarding the transformation distance. Hence, we adopt the top ETF features to facilitate their alignment.

for any class c , which is identical among all clients. During training, each client n uses these predetermined ETF prototypes as regularization to normalize its outputted features:

$$\mathcal{R}_g^n = \frac{1}{D_n d_s} \sum_{c \in \mathcal{Y}_n} \sum_{x_{n,c,i} \in \mathbb{D}_n^c} \|\mathbf{h}_{n,c,i}^{L_s} - \tilde{\mathbf{h}}_c^{L_s}\|_2^2, \quad (4)$$

where $\mathcal{Y}_n$ denotes the set of classes in client n , $\mathbb{D}_n^c$ is the local dataset for class c , and $x_{n,c,i}$ denotes the i -th sample with class c .

5.2 Align Personalized features with Local ETF

To enhance the separability of personalized features, we align local features $\mathbf{h}_{n,i}^{L-1} \in \mathbb{R}^d$ generated by the full extractor with local ETF prototypes. Similarly, for ease of alignment, each client needs to identify *matching* local ETF prototypes that are close to the local features transformed from global features by the personalized layers, as shown in Figure 6. This is accomplished by computing the mean of local features for each class $\bar{\mathbf{h}}_{n,c}^{L-1}$ and for all classes $\bar{\mathbf{h}}_n^{L-1}$, respectively. Each client then calculates the length average of normalized features $\text{Avg}_i \|\mathbf{h}_{n,i}^{L-1} - \bar{\mathbf{h}}_n^{L-1}\|$, and the normalized mean of features for each class, denoted $\hat{H}_n^{L-1} = [\hat{\mathbf{h}}_{n,1}^{L-1}, \dots, \hat{\mathbf{h}}_{n,C_n}^{L-1}]$. Here, $\hat{\mathbf{h}}_{n,c}^{L-1}$ is defined as $\frac{\bar{\mathbf{h}}_{n,c}^{L-1} - \bar{\mathbf{h}}_n^{L-1}}{\|\bar{\mathbf{h}}_{n,c}^{L-1} - \bar{\mathbf{h}}_n^{L-1}\|}$, and C_n represents the number of classes in the local dataset. Next, each client generates ETF vectors using the definition (2) while minimizing the transformation distance from local features to local ETF features, i.e., $\mathbf{E}_n = [\mathbf{e}_{n,1}, \dots, \mathbf{e}_{n,C_n}] \in \mathbb{R}^{d \times C_n}$ that approach the output feature means $\hat{H}_n^{L-1}$:

$$\begin{aligned} \min_{\mathbf{U}} \|\hat{H}_n^{L-1} - \mathbf{E}_n\|_2^2 &= \left\| \hat{H}_n^{L-1} - \sqrt{\frac{C_n}{C_n - 1}} \mathbf{U} (\mathbf{I}_{C_n} - \frac{1}{C_n} \mathbf{1}_{C_n} \mathbf{1}_{C_n}^T) \right\|_2^2, \\ \text{s.t., } \mathbf{U} \mathbf{U}^T &= \mathbf{I}, \end{aligned} \quad (5)$$

where $\mathbf{U} \in \mathbb{R}^{d \times C_n}$ is the rotation matrix. Equation (5) can be solved by the method in [53], where the computational complexity is merely $8dC_n^2 + O(C_n^3)$ given the small size of C_n . More discussions are in Appendix A.4. Like the computation of global prototypes in (3), each client constructs the local ETF prototypes using NC2:

$$\tilde{\mathbf{h}}_c^{L-1} = \text{Avg}_i \|\mathbf{h}_{n,i}^{L-1} - \bar{\mathbf{h}}_n^{L-1}\| \mathbf{e}_{n,c} + \bar{\mathbf{h}}_n^{L-1} \quad (6)$$

for each class $c \in \mathcal{Y}_n$. Then, each client n leverages these local ETF prototypes as regularization to train of personalized layers:

$$\mathcal{R}_p^n = \frac{1}{D_n d} \sum_{c \in \mathcal{Y}_n} \sum_{x_{n,c,i} \in \mathbb{D}_n^c} \|\mathbf{h}_{n,c,i}^{L-1} - \tilde{\mathbf{h}}_c^{L-1}\|_2^2. \quad (7)$$

Table 1: Comparison of test accuracy on the four datasets. The best result is bolded. Each experiment is conducted three times.

Method	Fashion-MNIST(%)	CIFAR-10 (%)	CIFAR-100 (%)		Tiny-ImageNet (%)	
Shards (n, S)	(100, 5)	(100, 5)	(20, 10)	(100, 10)	(20, 10)	(100, 10)
FedAvg-FT	95.12±0.21	79.77±0.42	64.45±0.45	56.59±0.51	57.19±0.86	44.25±0.71
Ditto [51]	90.13±1.41	63.21±1.03	60.32±1.22	33.66±1.58	53.29±1.33	37.42±1.55
FedPer [52]	94.22±0.62	76.86±0.71	63.47±0.47	47.58±0.63	55.94±0.74	39.08±0.87
FedBABU [25]	94.51±2.35	69.78±2.46	65.11±2.23	52.24±1.91	58.28±1.56	41.62±1.81
FedRep [31]	94.59±0.91	73.39±0.52	59.76±0.69	46.47±0.78	58.82±0.92	35.66±1.01
FEDAVG ETF [41]	86.43±1.14	59.37±1.54	55.74±1.23	32.42±1.72	51.09±1.21	33.09±1.32
FedETF [42]	95.85±1.12	79.39±1.34	65.17±1.64	56.98±1.21	60.23±1.32	49.78±1.04
AlignFed [38]	92.56±1.25	79.80±1.13	61.11±1.59	55.31±0.64	57.64±1.28	48.22±1.77
PartialFed [28]	94.62±1.33	78.56±1.94	58.37±1.62	56.24±1.29	58.72±1.83	41.64±1.37
FedCAC [36]	95.56±0.36	71.73±0.83	61.84±1.57	54.77±1.98	61.57±2.36	38.48±1.94
CD2pFed [39]	95.48±0.83	77.28±0.74	63.37±0.91	54.18±1.43	56.28±0.67	44.69±0.82
pFedHN [37]	90.38±1.56	79.77±1.76	60.08±1.08	55.86±1.13	55.95±0.71	45.81±1.81
FedPAC [32]	95.76±0.33	80.22±0.54	64.88±0.59	57.35±0.70	58.90±0.65	50.31±0.83
pFedDKS	96.02±0.46	81.35±0.39	67.55±0.66	58.23±0.82	63.19±0.77	54.09±0.61
Dirichlet (n, α)	(100, 0.1)	(100, 0.1)	(20, 0.05)	(100, 0.05)	(20, 0.05)	(100, 0.05)
FedAvg-FT	97.11±0.65	90.06±0.74	61.83±1.15	62.14±1.31	40.99±0.65	37.26±0.65
Ditto [51]	95.46±0.92	84.39±0.76	57.24±1.54	55.81±1.32	35.76±1.73	33.21±2.01
FedPer [52]	96.63±0.75	89.47±0.99	58.35±1.02	54.64±0.83	39.49±0.97	32.21±1.11
FedBABU [25]	96.53±1.92	90.39±1.33	61.62±2.35	59.98±2.12	39.46±2.40	32.35±2.27
FedRep [31]	96.66±0.34	88.98±0.27	57.57±1.05	53.45±0.89	39.42±0.88	29.98±0.97
FEDAVG ETF [41]	92.37±1.52	82.91±1.42	52.91±1.92	50.27±1.38	34.95±1.74	26.35±1.22
FedETF [42]	96.98±0.80	88.46±1.67	62.09±1.33	60.26±1.61	42.93±0.94	38.19±1.37
AlignFed [38]	95.85±0.68	89.97±1.46	58.12±1.83	56.03±0.96	40.36±1.76	39.92±1.01
PartialFed [28]	94.83±1.42	89.83±1.83	61.47±1.70	58.94±1.28	42.77±0.92	30.94±1.90
FedCAC [36]	97.10±0.62	88.15±0.24	60.05±1.02	59.42±1.55	42.75±1.36	40.37±0.79
CD2pFed [39]	96.79±1.13	89.07±0.68	58.79±1.26	60.77±1.08	41.38±1.47	39.68±1.15
pFedHN [37]	95.39±1.30	86.38±1.61	60.67±1.79	58.81±1.77	38.32±1.26	37.16±2.82
FedPAC [32]	97.19±0.32	90.40±0.44	62.21±1.22	61.52±0.87	43.27±0.59	37.25±0.78
pFedDKS	97.28±0.29	91.11±0.17	62.85±0.91	62.90±0.86	46.66±0.85	42.33±0.96

5.3 Joint Training Loss

pFedDKS is to join (1), (4), and (7) together:

$$\min_{\mathbf{W}} \mathcal{L}(\mathbf{W}) := \sum_{n=1}^N \frac{D_n}{D} \mathcal{L}_n(\mathbf{w}_n), \quad (8)$$

$$\mathcal{L}_n(\mathbf{w}_n) = \mathcal{L}_{CE}(\mathbf{w}_n) + \lambda_g \mathcal{R}_g^n(\mathbf{w}_n^{1:L_s}) + \lambda_p \mathcal{R}_p^n(\mathbf{w}_n^{L_s+1:L}),$$

where $\mathbf{w}_n^{1:L_s}$ refers to the 1 to L_s model layers in client n . Before using the joint training loss, we first conduct a *warmup* training stage, i.e., running several rounds of FedAvg across clients. Then, we compute the global and local ETF vectors using (3) and (6) separately, based which the regularizations $\mathcal{R}_g^n$ and $\mathcal{R}_p^n$ are obtained. After that, we adopt the generally adopted alternative optimization strategy [31, 32] to solve the joint loss (8). The workflow of solving (8) is shown in Appendix A.5.

5.4 Discussion and Analysis

Privacy. Our method only requires each client to upload the information of features once, specifically the average value of all classes' features, to calculate the global ETF, after which there is no need to upload any further feature-related information. Compared with other methods that require uploading the prototypes of all classes every round, our approach leads to significantly less privacy leakage. Afterwards, our method typically only necessitates

the interaction between client and server for exchanging partial extractor layers, sharing even less information than FedAvg.

Theoretical guarantee. We show that pFedDKS's converged features satisfy NC conditions, with maximized local feature separability indicating strong personalization. The proof is in Appendix A.6.

THEOREM 1. *When the objective (8) achieves its optima, the separability among local class features obtained by the personalized models reaches maximized, i.e., satisfying the conditions of NC1 and NC2.*

6 Evaluations

6.1 Experimental Setup

Datasets and model. We consider four popular datasets in experiments: Fashion-MNIST[54], CIFAR-10[55], CIFAR-100[55] and Tiny-ImageNet [56], which contains 10, 10, 100 and 200 classes respectively. A simple CNN and the ResNet-18 [57] are adopted. We use CNN in easy tasks such as Fashion-MNIST and CIFAR-10, while ResNet-18 in CIFAR-100 and Tiny-ImageNet. Details about models, decoupled layers, and warmup impacts are shown in Appendix B.

Data partition. To evaluate the performance of our work in a heterogeneous scenario, we specify two Non-IID data partition methods called Shards [1] and Dirichlet [58]. In the Shards setting, the sorted samples are shuffled into $n * S$ shards, and assigned to n

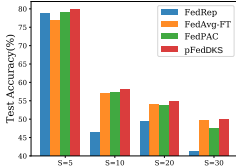
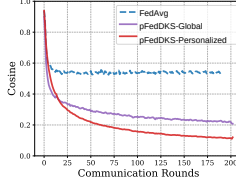
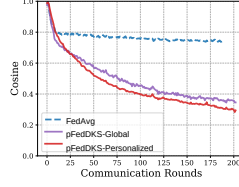


Figure 7: Varied heterogeneity on CIFAR-100.



(a) Fashion-MNIST



(b) CIFAR-10

Figure 8: Cosine similarity on two datasets.

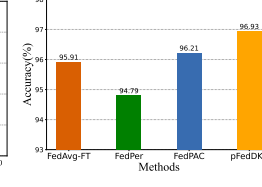


Figure 9: Comparison on News20 NLP dataset.

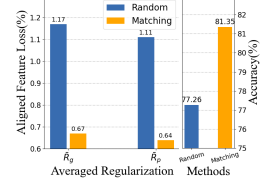


Figure 10: Random vs. matching ETF protos.

clients randomly. Each client owns an equal number of pieces. In the second setting, data distribution over clients satisfies the Dirichlet distribution with α characterizing the degree of heterogeneity.

Baselines. We compare against various pFL approaches: 1) methods that learn a global model as traditional FL algorithms including FEDAVG ETF [41]; 2) locally fine-tune the global models after training, including FedAvg-FT (fine-tuning), FedBABU [25], FedETF [42]; 3) methods with pre-defined regularization, such as Ditto [51]; 4) techniques with decoupled parameters, including FedCAC [36], FedPer [52], AlignFed [38], FedRep [31], CD2pFed [39], PartialFed [28], and FedPAC [32]; 5) others including pFedHN [37].

Implementation. We implement the whole experiment in a simulation environment based on PyTorch. To ensure the accuracy and repeatability of the experiments, we refer to a pFL benchmark [59] from the open-source community. We employ the mini-batch SGD optimizer with learning rate of 0.02 for all methods and datasets. The batch size is set to 32 unless explicitly specified. The number of local training epochs for pFedDKS is set to $E = 5$, and the personalized training epochs is $E_p = 5$ for CIFAR-100 and Tiny-ImageNet, $E_p = 3$ for Fashion-MNIST and CIFAR-10. The hyper-parameters $\lambda_g = \lambda_p = 10$ for all the settings. All the clients will join the aggregation for a total of 20 clients and we randomly select 20% clients per round for local training for a total of 100 clients.

6.2 Experiment Results

Performance comparison. The comparison results are shown in Table 1. In order to demonstrate the generalization of our method, we compare them using two different Non-IID settings, Shards and Dirichlet distribution. We apply different data distributions on different datasets and various numbers of clients. We can see that our proposed pFedDKS achieves the best performance in almost all settings, dominating the other methods on average test accuracy, which demonstrates the effectiveness and benefit of both alignments with pre-defined global ETF and local ETF. In addition, pFedDKS can make more improvements with more total clients. Specifically, pFedDKS is 7.6% better than the second-best baseline FedPAC with Shards and improves the accuracy by 9.2% than the second-best FedAvg-FT with Dirichlet on Tiny-ImageNet with 100 total clients. We can also notice that FedAvg with fine-tuning is a strong baseline for pFL, which is better than many specialized methods.

Table 2: Results on Cifar10 with various Dir(α) values.

Dir(α)	FedAvg-FT	FedPer	FedPAC	pFedDKS
0.1	90.06	89.47	90.40	91.11
1	73.36	65.31	74.19	75.38
10	65.05	53.81	66.62	67.89

Performance over various data heterogeneity. We vary the values of Shards(S) and Dir(α) to simulate different Non-IID levels.

Smaller values indicate stronger heterogeneity. We assess CIFAR-100 and CIFAR-10 with 100 clients, where Figure 7 and Table 2 present the results. pFedDKS consistently outperforms other baselines, evidencing its adaptability and robustness across diverse heterogeneous data settings. Despite increased local class counts and heightened classification complexity causing failure for methods like FedRep, pFedDKS maintains a distinct advantage.

Ablation studies. Our pFedDKS has three key regularization components, partial sharing (PS), partial global ETF alignment (GEA), and local ETF alignment (LEA). As LEA is based on GEA, we conduct ablation studies to verify the effectiveness of PS, GEA and the combination of them. The results on four datasets with 100 clients. As shown in Tab.3, we can find that PS and GEA are essential for performance improvement and the combination of the three components will achieve the most satisfactory model performance.

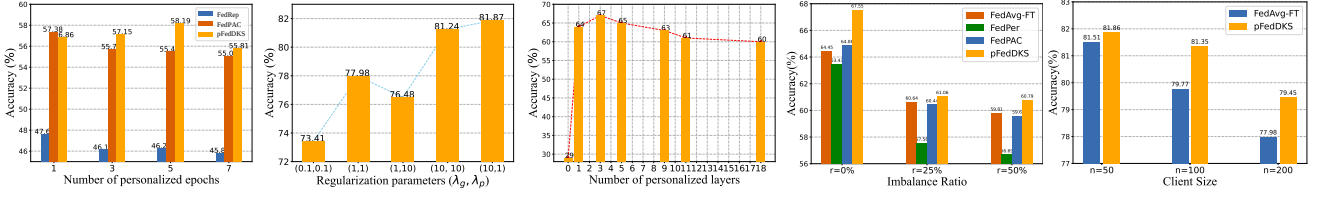
Table 3: Ablation study. 'w' denotes 'with'.

Method	FMNIST	CIFAR-10	CIFAR-100	Tiny-ImageNet
Shards (n, S)	(100, 5)	(100, 5)	(100, 10)	(100, 10)
FedAvg	89.85	63.27	23.78	9.65
w PS	92.39	72.98	45.37	28.11
w PS&GEA	95.71	80.42	55.63	51.14
pFedDKS	95.98	81.25	58.19	54.11

Feature separability. In order to further verify the effectiveness of our proposed global and local ETF alignment, we try to illustrate the feature separability among different classes. As shown in Figure 8, we plot the average cosine similarities between the feature centers of all classes during the training process. It can be seen that the feature similarity between classes after GEA is significantly lower than that of FedAvg, indicating that GEA can help the model better learn the difference between classes under the Non-IID setting. Meanwhile, feature discrepancy after personalized layers (deep extractor layers) further declines, meaning efficient adaptation of local data for personalization.

Evaluation over text datasets. To verify the effectiveness of our method on more data types, we compare our methods to baselines on the 20news text dataset with a two-layer LSTM as the model [60]. As shown in Figure 9, our method still performs the best.

Benefits of matching ETF prototypes. To verify the advantage of calculating ETF prototypes by matching real features, we compared them with ETF prototypes generated randomly. We mainly use $\bar{R}_g = R_g/d_s$ and $\bar{R}_p = R_p/d$ which are used to calculate the average norm distance between the real features and the ETF prototypes as the comparison metric. As shown in Figure 10, the matching ETF prototypes not only are closer to the real features, i.e., have smaller $\bar{R}_g$ and $\bar{R}_p$, but also result in higher accuracy.



(a) Impact of personal ep.s (b) Regularized parameters (c) Impact of split layers (d) Varied imbalance ratios (e) Impact of client size
Figure 11: Impact of hyperparameters.

Impact of learning rate. The results with various learning rates are shown in Table 4, which are conducted on CIFAR-10 and CIFAR-100 with the number of clients being 20. As can be seen, our algorithm consistently outperforms FedAvg-FT and FedPAC.

Table 4: Comparison of our method to high-performance baselines on different learning rates $\eta = 0.005, 0.01, 0.02$.

Method	CIFAR-10			CIFAR-100		
	0.005	0.01	0.02	0.005	0.01	0.02
FedAvg-FT	75.85	78.51	78.22	64.24	64.79	65.05
FedPAC	76.39	78.65	79.77	64.27	64.52	65.51
pFedDKS	78.06	82.30	81.25	66.65	66.57	67.55

Impact of personalized epoch. The results with various personalized epochs are demonstrated in Figure 11a, which were conducted on CIFAR-100. FedRep and FedPAC also need to perform personalized training on unshared classifiers. Obviously, pFedDKS needs more local training epochs because of more customized layers.

Sensitivity analysis. We conduct the sensitivity analysis on the weights of alignment terms, λ_g and λ_p . As expected, two parameters jointly regulate the global and local feature alignment process throughout the training process. As shown in Figure 11b, the two hyper-parameters should be set to a large value so that the global and local ETF alignment can significantly learn global knowledge as well as local personalized features.

Impact of different personalized layers. We conduct evaluations on CIFAR-100 with ResNet-18 by ranging the splitting layers to verify our statement. The results are shown in Figure 11c. As can be seen, fully sharing the feature extractor does not achieve the best performance ($x = 0$). Customizing 3 layers (i.e., 2 feature extractor layers and 1 classifier layer) receives maximal improvement. The observation also exhibits that even if partially sharing the feature extractor is necessary, sharing how many layers is still important.

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Impact of global data imbalance. We conduct experiments with global class imbalance on the CIFAR-100 (20,10) setting. We remove a proportion r of the data from the first 50 classes. As shown in Figure 11d, our method demonstrates a more significant advantage over the baseline when the class imbalance is severe.

Scalability. We run experiments on CIFAR10 by ranging the client number from 50 to 200. As shown in Figure 11e, the accuracy of

both methods decreases as the number of clients increases. This is because personalized models depend more on local data, and less local data leads to lower performance. However, our method is less affected by the increase in clients compared to FedAvgFT, and the gap widens as the client number grows beyond 50.

Table 5: Single-round communication cost of different methods. FedPer and pFedDKS split at 15-th layer of ResNet18.

Method	Communication Cost (MB)				Comm. Components
	10	30	50	100	
FedAvg-FT/Ditto	44.59	44.59	44.59	44.59	Full model
FedPer	24.63	24.63	24.63	24.63	Partial feature extractor
FedBABU	42.63	42.63	42.63	42.63	Full feature extractor
FedRep/FedPAC	42.65	42.69	42.73	42.83	Full extractor & Prototypes
pFedDKS	24.63	24.63	24.63	24.63	Partial feature extractor

Reduced communication cost. Compared to existing prototype-based methods, our method reduces communication costs by uploading fewer parameters and not uploading prototypes. The communication cost is computed based on the model size, the number of classes in each client, and number of shared layers. We present the single-round communication cost of each client by taking ResNet18 over ImageNet as an example, as shown in Table 5. It can be observed that pFedDKS communicates the least which is the same as FedPer but achieves the higher accuracy.

7 Conclusion

This paper dives into the principle of partial knowledge sharing in personalized FL. We introduce the theory of neural collapse to show that the feature extractor should not be fully shared to maximize the personalization performance. Specifically, we show that the separability of the optimal global features is smaller than the optimal personalized ones. Based on this finding, we propose a novel method pFedDKS which applies a novel detached feature prototype scheme over the partial sharing scheme. Detached global and local prototypes are generated based on the neural collapse conforming to the ETF structure such that the separability of global features and personalized features can be maximized respectively. Extensive experiments demonstrate the effectiveness of our proposed method.

Acknowledgments

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A Motivation

A.1 Average Strategy in Motivation

We adopt a strategy like FedRep. In this context, our motivation involves N_A major clients and N_B minor clients, with the count of each minority class being represented as D_{B_1} and D_{B_2} across two types of clients, respectively. The aggregation of each class c prototype $\bar{\mathbf{h}}_c$ follows a quantity-aware approach:

$$\bar{\mathbf{h}}_c = \sum_{n=1}^{N_A} \frac{D_{B_1}}{N_A D_{B_1} + N_B D_{B_2}} \bar{\mathbf{h}}_c^n + \sum_{n=N_B+1}^{N_A+N_B} \frac{D_{B_2}}{N_A D_{B_1} + N_B D_{B_2}} \bar{\mathbf{h}}_c^n \quad (9)$$

$$= \frac{1}{N_A D_{B_1} + N_B D_{B_2}} (N_A D_{B_1} \bar{\mathbf{h}}_c^1 + N_B D_{B_2} \bar{\mathbf{h}}_c^2) \quad (10)$$

where we view all major clients as having the identical prototype as $\bar{\mathbf{h}}_c^1$, and the same applies to all minor clients as $\bar{\mathbf{h}}_c^2$. This means that the global prototype approaches the local prototype of major clients $\bar{\mathbf{h}}_c \rightarrow \bar{\mathbf{h}}_c^1$ as $N_A D_{B_1} / (N_B D_{B_2}) \rightarrow \infty$, i.e., $N_A / N_B \rightarrow \infty$ as D_{B_1} is close to D_{B_2} . Due to the diminished separability of major clients, the separability of aggregated prototypes becomes smaller than that of local prototypes within minor clients, signaling the presence of interference.

A.2 Proof of Proposition 1

PROOF. According to NC1, NC3, and minority collapse, the features of minority classes in each of the N_A majority clients will converge to a single vector denoted as $\bar{\mathbf{h}}_{n,c}^{L-1} - \bar{\mathbf{h}}_A^{L-1} \rightarrow 0$, for all $1 \leq n \leq N_A$, $C_A < c \leq C$, as the ratio $D^A/D^B \rightarrow \infty$. Meanwhile, the N_B minority clients are expected to learn features $\bar{\mathbf{h}}_{n,c}^{L-1}$ that satisfy the ETF condition, which maximize the separability between classes $C_A < c \leq c' \leq C$. Because all clients share features, the separability of minority classes will be further reduced due to interference from the majority clients. This means that as $\frac{N_A D^B}{N_B D^M} \rightarrow \infty$, we have $\frac{1}{N} \sum_{n=1}^N \bar{\mathbf{h}}_{n,c}^{L-1} - \bar{\mathbf{h}}_A^{L-1} \rightarrow 0$ for all $C_A < c \leq C$. $\square$

A.3 Proof of Proposition 2

PROOF. Consider the global model has achieved optimal, where the feature formulates the ETF structure. Then, based on the definition of ETF vectors in (2), the angle between the feature means of two classes is $\arccos(-\frac{1}{C-1})$. The angle between the mean values of the features output by the locally optimal model for the two local classes is $\arccos(-\frac{1}{C_n-1})$. Therefore, the difference between the angle of the global model and that of the local model is $\arccos(-\frac{1}{C_n-1}) - \arccos(-\frac{1}{C-1})$. $\square$

A.4 Complexity of solving equation (5)

In the following text, we present the key derivations of [53] for solving the problem (5). Considering an optimization problem with

the matrix orthogonality constraints:

$$\min_{X \in \mathbb{R}^{n \times p}} \mathcal{F}(X), \quad \text{s.t. } X^T X = I, \quad (11)$$

Given a feasible point X and the gradient $G := D\mathcal{F}(X) = \left(\frac{\partial \mathcal{F}(X)}{\partial X_{ij}} \right)$, we define a skew-symmetric matrix A as either

$$A := GX^T - XG^T \quad \text{or} \quad (12)$$

$$A := (P_X G)X^T - X(P_X G)^T, \quad \text{where } P_X := \left(I - \frac{1}{2}XX^T \right). \quad (13)$$

The new trial point is determined by the Crank-Nicolson-like scheme

$$Y(\tau) = X - \frac{\tau}{2} A (X + Y(\tau)), \quad (14)$$

where $Y(\tau)$ is given in the closed form:

$$Y(\tau) = QX, \quad \text{where } Q := \left(I + \frac{\tau}{2}A \right)^{-1} \left(I - \frac{\tau}{2}A \right). \quad (15)$$

Known as the Cayley transformation, Q leads to several nice properties: the curve $Y(\tau)$ is smooth in τ , $Y(0) = X$, and most importantly $(Y(\tau))^T Y(\tau) = X^T X$ for all $\tau \in \mathbb{R}$.

Equation (15) requires inverting $(I + \frac{\tau}{2}A)$ in order to preserve the orthogonality constraints. It is often numerically cheaper than computing SVDs or geodesics. When $p = 1$, the inverse $(I + \frac{\tau}{2}A)^{-1}$ (and thus $Y(\tau)$) is given in a closed form with a few vector-vector products. When p is much smaller than $n/2$ —which is the case in many applications— A has rank $2p$; hence it follows from the Sherman–Morrison–Woodbury (SMW) theorem that one only needs to invert a smaller $2p \times 2p$ matrix. The total flops for computing $Y(\tau)$ is $8np^2 + O(p^3)$, and updating $Y(\tau)$ for a different τ needs $4np^2 + O(p^3)$ flops. Considering $p = C_n$ in the problem (5), the computational complexity is therefore $8dC_n^2 + O(C_n^3)$.

A.5 Algorithm

The algorithm workflow of pFedDKS is present in Algorithm 1.

A.6 Proof of Theorem 1

PROOF. Our primary goal is to satisfy the second condition, NC2, which aims to make the local class features converge to an ETF structure to maximize their separability, thereby expanding the classification boundaries, rather than fulfilling all NC conditions. Of course, we can also demonstrate that when Eq. (7) reaches its optimal value, the trained features will also meet the NC1 condition. The proofs for both conditions are as follows:

NC1: When Eq. (7) reaches its optimal value, for any sample $(x_{n,i}, y_{n,i})$ with $y_{n,i} = c$, its feature $\mathbf{h}_{n,c,i}^{L-1}$ will converge to the ETF prototype $\tilde{\mathbf{h}}_c^{L-1}$ corresponding to class c , i.e., $\mathbf{h}_{n,c,i}^{L-1} = \tilde{\mathbf{h}}_c^{L-1}$. Therefore, it is evident that for all samples belonging to a local class c , the following conclusion can be drawn:

$$\text{Avg}_i \mathbf{h}_{n,c,i}^{L-1} = \text{Avg}_i \tilde{\mathbf{h}}_c^{L-1} = \tilde{\mathbf{h}}_c^{L-1}$$

Moreover, since the features of all samples within each class are identical, the variance of the features within the same class is naturally zero. Hence, when Eq. (7) reaches its optimal value, the NC1 condition is satisfied.

NC2: Because $\tilde{\mathbf{h}}_c^{L-1}$, $c = 1, \dots, C_n$ are constructed based on Definition 1 of the ETF, which is solved by Eq. (5) and Eq. (6), they satisfy the ETF structure. Additionally, from the proof of NC1, we know

Algorithm 1 Workflow of pFedDKS

Input: Dataset $\mathbb{D}_n$, model $\mathbf{w}_n$, $n = 1, \dots, N$, regularization parameter λ , learning rate η , and total rounds T .

Output: Trained personalized models $\mathbf{w} = [\mathbf{w}_1, \dots, \mathbf{w}_N]$.

```

1: Initialize the model parameters  $\mathbf{w}_0$ ;
2: procedure SERVER-SIDE OPTIMIZATION
3:   Distribute  $\mathbf{w}_0$  to all clients;
4:   for each communication round  $t \in \{1, 2, \dots, T\}$  do
5:     Randomly select a set of clients  $\mathcal{S}_t$ ;
6:     for each selected client  $n \in \mathcal{S}_t$  in parallel do
7:       Distribute  $\mathbf{w}_n^{1:L_s}$  to all selected clients;
8:        $\mathbf{w}_n^{1:L_s} = \text{ClientLocalUpdate}(\mathbf{w}_n^{1:L_s})$ ;
9:     Aggregate shared layers  $\mathbf{w}^{1:L_s} = \sum_{n \in \mathcal{S}} \frac{D_n}{\sum_{i \in \mathcal{S}} D_i} \mathbf{w}_n^{1:L_s}$ ;
10:  procedure CLIENTLOCALUPDATE
11:    Client  $n$  receives  $\mathbf{w}_n^{1:L_s}$  from the server;
12:    for each local epoch  $e$  from 1 to  $E_1$  do
13:      for mini-batch  $\xi_n \subseteq \mathbb{D}_n$  do
14:        Update personalized layers on local data:  $\mathbf{w}_n^{L_s+1:L} =$ 
 $\mathbf{w}_n^{L_s+1:L} - \eta \nabla_{\mathbf{w}_n^{L_s+1:L}} \mathcal{L}_{CE}(\mathbf{w}_n; \xi_n) - \eta \lambda \rho \nabla_{\mathbf{w}_n^{L_s+1:L}} \mathcal{R}_p^n(\mathbf{w}_n^{L_s+1:L})$ ;
15:      for each local epoch  $e$  from 1 to  $E_2$  do
16:        for mini-batch  $\xi_n \subseteq \mathbb{D}_n$  do
17:          Update shared model parameters:  $\mathbf{w}_n^{1:L_s} = \mathbf{w}_n^{1:L_s} -$ 
 $\eta \nabla_{\mathbf{w}_n^{1:L_s}} \mathcal{L}_{CE}(\mathbf{w}_n; \xi_n) - \eta \lambda g \nabla_{\mathbf{w}_n^{1:L_s}} \mathcal{R}_g^n(\mathbf{w}_n^{1:L_s})$ ;
          return local parameters  $\mathbf{w}_{t+1}$ ;
```

that the average of the features for each class converges to the corresponding ETF prototype $\tilde{\mathbf{h}}_c^{L-1}$, satisfying the NC2 condition. $\square$

B Experimental Details

B.1 Dataset and Partition

We adopt four datasets with various class numbers and image resolutions: (1) Fashion-MNIST which contains 10 classes of 28×28 grayscale images, each class has 6000 images for training and 1000 images for testing; (2) CIFAR-10 with 10 classes of 32×32 color images, each class has 5000 images in the training dataset and 1000 images in the test dataset; (3) CIFAR-100, which contains 100 classes of 32×32 images, each class has 500 images and 100 images for training and testing, respectively; (4) Tiny-ImageNet which contains 64×64 resolution images with 200 classes, and each class has 500 and 50 images in the training and testing datasets.

We consider 2 data partition methods to simulate the non-i.i.d. case in the real world: (1) Shards, which divide the sorted samples into $n * S$ shards, and assigned these shards to n clients randomly. Obviously, each client contains S pieces of samples, but that doesn't mean each client owns S classes of samples. A client may be assigned shards that belong to the same label, which results in a slightly uneven distribution of local samples. (2) Dirichlet, a more practical setting, assumes that the samples among clients satisfy a specific distribution determined by parameter α .

B.2 Model Architecture and Decouple Method

Similar to [40], We conduct the CNN with 2 convolution layers and 3 linear layers. Following FedRep [31] and FedPAC [32], we

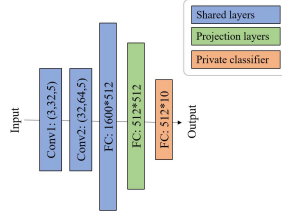


Figure 12: pFedDKS-CNN architecture

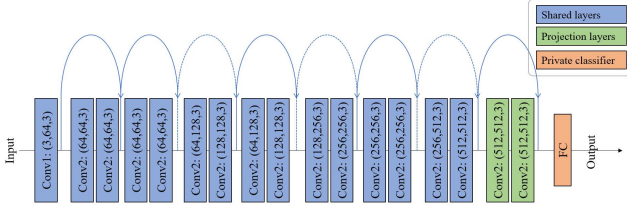


Figure 13: pFedDKS ResNet-18 architecture

decouple the head of CNN during training and aggregating to allow personalized classification. Furthermore, the last linear layer before the classifier in pFedDKS -CNN is split as another private layer for personalized ETF prototype projection. It is worth noting that the dimension of the shared fully connected layer in the figure is based on the result that the input data is CIFAR-10. For Fashion-MNIST, the dimension of the output features of shared backbone layers should be 1024 instead of 1600. The structure of pFedDKS -CNN is shown in Figure 12.

For ResNet-18, half layers of the last basic block are split as personalized projection layers, as shown in Figure 13. We align the shared layers feature with global ETF vectors and constraint the projection layers features with optimal personalized ETF prototype. The batch normalization layers are not adopted in our experiment because of the potential performance penalty in federated learning settings. One solution is to set batch normalization layers personalized [61], which is orthogonal with our method. The rest layers of the network remains unchanged.

B.3 Implementation Details

We employ the mini-batch SGD optimizer with a learning rate of 0.02 for all methods and datasets. The weight decay is set to $5e-4$ and momentum is set to 0.5 for all baselines and pFedDKS. The batch size is set to 32 unless explicitly specified, and the training epoch of backbone layers for all partial learning methods is set to 5. The personalized training epoch is $E_p = 5$ for CIFAR-100 and Tiny-ImageNet, $E_p = 3$ for Fashion-MNIST and CIFAR-10. All the clients will join the aggregation for a total of 20 clients and we randomly select 20% clients per round for local training for a total of 100 clients. The hyper-parameters are set to $\lambda_g = \lambda_p = 10$ for pFedDKS. For FedPAC, we set $\lambda = 1$ and $E_p = 1$ according to the original literature. For Ditto, we set $\lambda = 1$ for all settings by searching over $\{0.1, 1, 10\}$. For a fair comparison, the personalized training epoch of FedRep and FedPer is set the same as pFedDKS. The fine-tune steps of FedBABU and FedAvg are 5.

Table 6: The grid search results of different methods.

FMNIST	$\eta = 0.005$	$\eta = 0.01$	$\eta = 0.02$
FedAvg-FT	94.51	95.27	95.75
FedPAC($\lambda = 0.1$)	94.11	94.89	95.28
FedPAC($\lambda = 1$)	94.57	95.45	95.66
FedPAC($\lambda = 10$)	95.04	95.44	95.90
pFedDKS	94.91	95.62	95.98

Table 7: The grid search results of different methods.

CIFAR-10	$\eta = 0.005$	$\eta = 0.01$	$\eta = 0.02$
FedAvg-FT	75.85	78.51	72.22
FedPAC($\lambda = 0.1$)	76.39	78.65	79.77
FedPAC($\lambda = 1$)	79.04	80.14	80.15
FedPAC($\lambda = 10$)	72.68	76.76	79.92
pFedDKS	78.06	82.30	81.25

B.4 Comparison via experiments with grid-searched hyper-parameters

To compare the best result of each method of different hyper-parameters. We range the learning rate of all methods from 0.005, 0.01 to 0.02. The hyper-parameter λ of FedPAC ranges from 0.1, 1 to 10. The hyper-parameter λ of our method remains unchanged, i.e., $\lambda = 10$. All methods run 200 rounds and the experimental results are presented in the following Table 6, 7, 8. It can be observed that our method still performs the best. Besides, we also find that a learning rate of 0.02 has little impact over the FedPAC as its hyper-parameter $\lambda = 1$ which is generally adopted by its original paper. Therefore, we mainly compare with it by running it with configurations $\eta = 0.02, \lambda = 1$.

Table 8: The grid search result of different methods.

CIFAR-100	$\eta = 0.005$	$\eta = 0.01$	$\eta = 0.02$
FedAvg-FT	64.24	64.79	65.05
FedPAC($\lambda = 0.1$)	64.27	64.52	65.51
FedPAC($\lambda = 1$)	63.38	63.37	63.61
FedPAC($\lambda = 10$)	59.44	57.40	62.78
pFedDKS	66.65	66.57	67.55

Table 9: Impact of Warmup

Warmup	0	10	20
pFedDKS	77.43	81.36	81.65

B.5 Impact of Warmup

In fact, the convergence curves of many Federated Learning papers show that convergence is very rapid in the early stages and gradually slows down later. Therefore, in our experiments, we generally run FedRep for the first 10 rounds before starting our method. We compared the use of warmup with no warmup on the Cifar10 dataset, and the results are shown in Table 9 below. It can be seen that warmup significantly improves the model training accuracy.